

Question ID: 1ce51655

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

A bowl contains **20** ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by **1** ounce every **4** days. If **9** ounces of water remain in this bowl, for how many days has it been uncovered?

Answer

- A. **3**
- B. **7**
- C. **36**
- D. **44**

Correct Answer: D

Rationale

Choice D is correct. It’s given that the bowl starts with **20** ounces of water and has **9** ounces of water remaining after a period of time has passed. The amount of water the bowl has lost during the time period can be found by subtracting the remaining amount of water from the amount of water the bowl starts with, which yields **20 — 9** ounces, or **11** ounces. This means the bowl loses **11** ounces of water during that period of time. It’s given that the amount of water decreases by **1** ounce every **4** days. Letting *t* represent the number of days the bowl has been uncovered, it follows that $\frac{1}{4} = \frac{11}{t}$. Multiplying both sides of this equation by **4t** yields *t* = **44**. Therefore, the bowl has been uncovered for **44** days.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of *t* for the equation $\frac{1}{4} = \frac{9}{t}$, not $\frac{1}{4} = \frac{11}{t}$.